

TORONTO BILLY BISHOP, ON, Canada

WMO: 712650

Lat: 43.6275N

Lon: 79.3966W

Elev: 77

StdP: 100.41

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -16.0	(c) -13.3	(d) -23.0	(e) 0.5	(f) -15.5	(g) -19.6	(h) 0.7	(i) -12.4	(j) 16.1	(k) -4.2	(l) 14.9	(m) -4.8	(n) 5.7	(o) 320	(p) 0.574

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 6.5	(c) 28.3	(d) 21.6	(e) 26.8	(f) 21.1	(g) 25.4	(h) 20.6	(i) 23.4	(j) 26.3	(k) 22.5	(l) 25.3	(m) 21.7	(n) 24.3	(o) 5.2	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	17.3	25.1	21.6	16.4	24.3	20.8	15.6	23.6	70.1	26.3	66.7	25.3	63.7	24.4	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.3	11.7	10.3	-19.4	32.1	3.3	2.2	-21.8	33.7	-23.8	35.0	-25.6	36.2	-28.1	37.8
			-20.1	25.1	3.2	1.2	-22.4	26.0	-24.2	26.7	-26.0	27.4	-28.3	28.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	-3.5	-2.7	0.9	6.5	12.5	18.1	21.0	21.1	17.6	10.8	4.8	-0.1
	DBStd	9.64	5.56	4.92	4.69	3.64	3.52	3.15	2.81	2.52	3.44	3.77	3.87	4.50
	HDD10.0	1689	418	357	282	113	13	0	0	0	1	36	158	313
	HDD18.3	3660	676	590	539	354	183	42	7	5	53	234	405	571
	CDD10.0	1326	0	0	1	9	92	243	342	346	229	61	3	0
	CDD18.3	255	0	0	0	0	4	35	91	93	32	1	0	0
	CDH23.3	931	0	0	0	0	18	121	396	312	82	2	0	0
	CDH26.7	158	0	0	0	0	2	22	82	43	8	1	0	0
Wind	WSAvg	4.8	6.1	5.9	5.3	5.2	4.3	3.8	3.5	3.6	3.9	4.7	5.1	5.8
Precipitation	PrecAvg	847	61	56	65	69	72	69	74	82	77	63	78	76
	PrecMax	1158	123	98	155	125	150	151	142	196	222	142	190	154
	PrecMin	648	21	20	13	30	26	17	20	17	10	12	13	24
	PrecStd	119	27	26	32	24	30	33	31	44	45	32	36	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.7	8.9	14.1	18.5	25.9	29.6	31.7	29.9	27.8	22.0	14.8	10.2
		MCWB	7.8	5.9	10.1	11.7	17.9	20.8	22.5	22.2	22.3	18.1	12.0	8.3
	2%	DB	6.7	6.1	10.5	15.2	22.2	26.4	28.8	27.8	25.5	19.6	12.6	8.2
		MCWB	5.6	4.2	7.4	9.8	16.9	20.0	21.5	22.0	21.1	16.5	10.5	6.9
	5%	DB	5.0	4.5	8.5	13.1	19.8	24.4	27.0	26.2	23.9	18.0	11.2	6.8
		MCWB	3.9	3.0	5.9	8.8	15.3	19.3	21.3	21.3	20.3	15.3	9.4	5.4
	10%	DB	3.6	3.3	6.7	11.4	17.8	22.8	25.4	25.0	22.7	16.5	10.0	5.5
		MCWB	2.6	1.9	4.6	7.7	13.8	18.7	20.6	20.8	19.6	14.2	8.3	4.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	6.8	11.0	13.4	19.9	23.2	24.6	24.5	23.3	19.4	13.0	8.9
		MCDB	8.6	8.5	12.9	16.3	23.5	25.8	28.7	27.2	26.3	21.1	14.0	9.8
	2%	WB	5.8	4.5	8.2	11.2	17.8	21.7	23.4	23.4	22.1	17.2	10.9	7.1
		MCDB	6.3	5.6	9.9	13.8	21.2	24.6	26.7	26.1	24.5	19.0	11.9	7.9
	5%	WB	4.1	3.2	6.3	9.5	15.8	20.5	22.4	22.5	21.1	15.9	9.7	5.7
		MCDB	4.8	4.4	7.9	12.2	18.9	23.2	25.3	25.1	23.2	17.5	10.9	6.6
	10%	WB	2.7	2.0	4.7	8.1	14.4	19.3	21.5	21.7	20.1	14.5	8.5	4.3
		MCDB	3.5	3.1	6.5	10.8	17.1	22.0	24.2	24.1	22.2	16.1	9.8	5.3
Mean Daily Temperature Range	5% DB	MDBR	6.0	5.9	6.1	6.6	7.4	7.3	7.5	6.5	6.6	5.8	5.1	5.0
		MCDBR	5.9	6.4	8.2	9.8	10.9	9.6	9.6	7.9	7.6	7.5	6.6	5.6
		MCWBR	5.8	5.7	6.1	6.3	6.5	5.3	5.1	4.7	5.0	5.6	5.7	5.1
	5% WB	MCDBR	5.9	6.5	7.8	9.0	10.0	8.4	8.2	6.7	6.8	6.6	6.1	5.5
		MCWBR	6.0	6.0	6.4	6.7	6.7	5.2	4.8	4.5	4.9	5.4	5.7	5.5
Clear-Sky Solar Irradiance	taub	0.303	0.324	0.337	0.384	0.414	0.443	0.437	0.423	0.390	0.368	0.322	0.304	
	taud	2.404	2.339	2.395	2.296	2.266	2.215	2.258	2.286	2.356	2.389	2.482	2.442	
	Ebn at Noon	826	865	903	880	860	832	833	832	834	804	799	788	
	Edh at Noon	76	96	104	124	132	140	133	125	108	92	71	68	
All-Sky Solar Radiation	RadAvg	1.41	2.15	3.43	4.43	5.38	5.84	5.86	5.16	4.13	2.48	1.55	1.13	
	RadStd	0.13	0.19	0.34	0.47	0.57	0.43	0.43	0.26	0.38	0.22	0.20	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page